

Technical Summary

Onboard Telemetry Triage and Prioritization for Satellite Mission Operations

1. Background & Motivation

Modern satellites, particularly small satellites and emerging constellations, generate increasingly dense telemetry streams while operating under persistent constraints in downlink bandwidth, ground contact windows, and operator attention. In current operations, telemetry is often downlinked in bulk and assessed on the ground using static thresholds or manual inspection, resulting in:

- Late anomaly awareness
- Inefficient use of downlink capacity
- Increased operational workload and cost

Operators frequently suspect that a significant portion of routine telemetry has low immediate operational value, but lack safe mechanisms to reduce or defer downlink without increasing mission risk.

2. Problem Statement

The core problem addressed is **how to prioritize telemetry data before downlink**, under realistic onboard constraints, while preserving operator trust and mission safety.

Specifically:

- Can onboard or near-onboard processing assist in **early classification of telemetry importance**?
 - Can this be achieved without autonomous decision-making or opaque machine-learning behavior?
 - Can the approach remain explainable, configurable, and compatible with certification-driven mission environments?
-

3. Proposed Concept

The concept explores a **lightweight onboard telemetry intelligence layer** that performs *preliminary triage* of telemetry streams prior to downlink.

Rather than deciding actions autonomously, the system classifies telemetry samples or windows into **operational priority categories**, for example:

- **Must-downlink immediately** (potential safety-critical or anomalous behavior)
- **Can-wait** (nominal or slowly varying conditions)
- **Can-discard or compress with audit trail** (highly repetitive or low-information content)

The output is intended to *inform downlink prioritization*, not to replace ground-based analysis or human decision-making.

4. Technical Approach (High-Level)

4.1 Data Scope

Initial focus is on **platform telemetry**, such as:

- Power system proxies (bus voltage, current stability)
- Thermal indicators
- Attitude or control error proxies

Payload data is explicitly *out of scope* in the initial phase.

4.2 Processing Strategy

The approach favors **simple, explainable techniques** suitable for constrained onboard environments:

- Rolling statistical baselines
- Adaptive thresholding (e.g. windowed z-score variants)
- Multi-channel consistency checks
- Trend and drift detection

These techniques are chosen to:

- Minimize computational load
- Avoid black-box inference
- Provide interpretable outputs to operators

Machine-learning components, if used, are limited to **lightweight pattern consistency checks**, not autonomous control or decision enforcement.

4.3 Human-in-the-Loop Design

A key design constraint is **non-autonomy**:

- The system does not execute corrective actions
- All decisions remain auditable and reversible
- Operators retain final authority

Classification outputs are accompanied by:

- Confidence indicators

- Contextual metadata (e.g. deviation magnitude, duration)
 - Optional raw data snapshots for verification
-

5. Validation Approach

A proof-of-concept has been developed using:

- Simulated multi-channel telemetry streams
- Controlled anomaly injection (spikes, drifts, cross-channel inconsistencies)

The objective is not flight qualification, but **concept validation**:

- Does early triage reduce unnecessary downlink volume?
 - Are anomalies flagged earlier than ground-only monitoring?
 - Are false positives manageable under conservative thresholds?
-

6. Intended Application Context

The concept is primarily targeted at:

- Small satellite missions
- Early constellation operators
- Cost-sensitive missions with limited ground resources

It is not positioned as a replacement for:

- Existing flight software
- Mission control systems
- Payload-level onboard analytics

Instead, it is envisioned as a **complementary decision-support layer** focused on telemetry relevance.

7. Key Assumptions & Open Questions

The work explicitly acknowledges open technical questions, including:

- What minimum onboard resources are realistically available across mission classes?
- Which telemetry types benefit most from early triage versus ground-only analysis?
- How conservative must thresholds be to remain acceptable to operators and certification bodies?

- How should auditability be implemented to maintain trust when data is deprioritized?

Feedback on these assumptions is a primary motivation for seeking expert input.

8. Scope Boundaries

To avoid overreach, the concept **does not** assume:

- Autonomous fault management
- Onboard control authority
- In-orbit data centers or large-scale compute

The focus remains on **decision support under constraint**, not autonomy.

9. Purpose of This Consultation

This summary is shared to seek **technical perspective**, specifically:

- Does this problem framing reflect real operational pain points?
- Is the scope realistic for onboard or edge execution?
- Are there mission classes where this approach would be particularly relevant or unsuitable?

Any high-level feedback, critiques, or boundary conditions would be greatly appreciated.

Author:

Dr. Rattapol Kasemrat

(early-stage concept; not flight-qualified)